WEEK 1 · DAY 1

Today, Wednesday 4 November

Two sessions in room E2 · 13:00–16:30

T. Santos, L. Batista · Ship and Ocean Systems Design 2026

The Plan of the Day

W1.1 13:00–15:00 · E2

The course and the project

- **The course**
  Objectives, evaluation, calendar, course site
- **The Ship Design Project**
  Two Baseline ships, D1–D10; start forming groups
- **Ship types 1.1**
  How ships are classified · 10 min
- **Activity 1.1 · 15 min**
  Find ships in StatCode 5, on your phone

W1.2 15:00–16:30 · E2

From the brief to the design process

- **Ship types 1.2**
  Deadweight, volume and linear-dimension carriers
- **Lecture 1.2 · Owner’s requirements**
  Who buys the ship and what the owner specifies
- **Activity 1.2 · in pairs · 20 min**
  Does the reference ship’s brief add up?
- **Lecture 1.3 · The design process**
  Life of a ship, the design spiral, phases A to D

Bring Today, and Prepare for Friday

Today

- Phone, pen and a calculator
- No computer needed today
- Activities are individual or in pairs, not graded

Friday 6 November

- Bring a laptop: exercises in Python or Excel
- Project groups are final
- Prepare the book sections below

Read before Friday · Papanikolaou (2026)

W1.3 · 13:30 · Main dimensions

- §1.3.6–1.3.7: dimensioning by ship category
- §2.1–2.3: displacement and main dimensions
- Appendix A: notes, A.1, A.3 and A.4

W1.4 · 15:30 · Displacement equation

- §2.13: the design equation
- Appendix C.1–C.2: Normand’s method

Consolidate Today: Extended Reading

Recommended after class, on the course site. Extended reading, not for evaluation.

W1.1 **Reading 1.1**

Ship Classification

How IMO, the EU and StatCode 5 each define ship types

W1.2 **Reading 1.2**

Who Buys the Ship: Real Cases

Five real deals: shipping line, tonnage provider, tramp owner, bareboat and leaseback

W1.2 **Reading 1.2B**

Who Owns the World Fleet

The top ten owning countries and their companies

Course site

l-u-i-s-b-a-t-i-s-t-a.github.io/PNSO-2026

All materials, session by session · office hours at calendly.com/vnsd-luisbatista/meetme